

ONEABILITY AI

Product Requirements Document (PRD) + System Architecture + Technology Stack

"One Platform. Every Ability. Independent Access."

Project Classification

Accessibility-first simulated digital payment assistant for academic demonstration. The current project demonstrates accessible payment interactions, QR parsing, AI-assisted intent understanding, safety checks, and secure authorization patterns without real money settlement.

Prepared in English from the supplied OneAbility AI feature summary and PRD documents.

Document Status and Source Alignment

This document consolidates the user-provided OneAbility AI project description with two supplied project documents: the current integrated-features summary and the Tanglish PRD planning document. Where the product vision is broader than the confirmed implementation, this report explicitly separates "current implementation" from "planned or aspirational capability" instead of treating them as the same thing.

Critical Scope Boundary

The current OneAbility AI build is a college mini-project simulation. It does not perform real UPI settlement and must not collect, store, transmit, log, or speak a real UPI PIN, OTP, or raw biometric data. Payment execution in this report therefore means simulated or mock execution unless a future regulated integration is explicitly described.

Source	Role in this report
OneAbility_AI_Current_Features_Tanglish(5).pdf	Primary basis for what is currently integrated, current stack, current persistence model, testing status, and confirmed project boundaries.
One_Ability_AI_PRD_Tanglish_Antigravity(5).pdf	Primary basis for product goals, functional requirements, accessibility requirements, security requirements, acceptance criteria, and MVP scope.
User-provided project description in this conversation	Adds the product pitch, seven highlighted features, intended multilingual voice experience, and a concise technology summary. Items that differ from confirmed implementation are marked accordingly.

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1. Executive Summary

OneAbility AI is an accessibility-first simulated digital payment assistant designed to let people with different accessibility needs complete common payment tasks through more than one interaction method. The product combines a modern payment-style interface with voice input, screen-reader semantics, captions, haptic cues, large touch targets, high contrast, keyboard support, dwell-click support, and simplified cognitive flows.

The distinguishing product principle is multimodal independence: a critical payment step should not depend only on sight, hearing, or precise touch. A blind user can rely on spoken feedback and screen-reader semantics; a deaf or hard-of-hearing user can rely on visible confirmations, captions and vibration; a motor-impaired user can use large targets, keyboard navigation and dwell interaction; and an elderly or cognitively overloaded user can use simplified, step-by-step flows.

The current implementation supports simulated Voice Pay, Scan & Pay, Pay Contact, Pay UPI ID, bank-linking demonstration, WebAuthn-based secure authorization where supported, beneficiary management, transaction history, accessibility settings, PWA shell caching, fraud-warning hooks, and Gemini-backed natural-language understanding for Tamil, English and Tanglish. Real UPI settlement, production KYC, real OTP-based bank verification, a cloud database, and real financial authorization remain outside the current mini-project scope.

2. Product Overview

Product name: OneAbility AI. AI assistant name: Dex. The application is designed as a universal-access payment experience rather than a disability-specific single-mode tool. It retains familiar payment concepts such as Scan & Pay, contacts, balance, bank selection, transaction history, bills and recharge, while adding accessible alternatives for each critical interaction.

One-line Pitch

OneAbility AI is an accessibility-first simulated digital payment assistant that enables Tamil, English and Tanglish interaction, QR-based payment simulation, adaptive accessibility modes, explicit confirmation, safety checks, and AI-assisted natural-language guidance.

2.1 Product Value Proposition

- Independent payment interaction for users who cannot rely on visual-only interfaces.
- Natural-language payment initiation in Tamil, English and Tanglish.
- Explicit read-back and confirmation before any simulated debit action.
- Accessible feedback through text, audio and haptic channels.
- Safety warnings for wrong receivers, suspicious QR payloads, duplicate attempts and unusually high amounts.
- Academic demonstration of secure design boundaries without handling real financial credentials.

3. Problem Statement and Opportunity

Mainstream digital payment applications are often usable with platform accessibility features, but complex navigation, small visual controls, QR alignment, audio-only prompts, and confirmation-heavy flows can still create friction for users with vision, hearing, motor or cognitive accessibility needs.

OneAbility AI focuses on demonstrating how a payment experience can expose the same intent through multiple accessible channels.

User need	Observed friction	OneAbility AI response
Blind / low-vision	Receiver, amount, QR details and result may be difficult to verify visually.	Read-aloud feedback, screen-reader semantics, high contrast, large text, voice-guided flow.
Deaf / hard-of-hearing	Audio-only alerts do not communicate state.	Captions, visible confirmation, flash-style alerts and vibration where supported.
Motor impairment	Precise tapping, repeated typing and swipe gestures may be difficult.	Large touch targets, keyboard navigation, dwell-click support and fewer taps.
Cognitive / elderly	Dense screens, jargon and too many choices increase error risk.	Simple wording, reduced visual noise, step-by-step confirmations.
Multilingual users	Rigid command syntax can make voice control unusable.	Natural Tamil, English and Tanglish intent understanding with safe field extraction.
Safety-sensitive users	Wrong receiver, suspicious QR or duplicate payment may be missed.	Explicit warnings, read-back confirmation and deterministic payment guards.

4. Product Vision, Goals and Non-Goals

4.1 Vision

A user should be able to complete a critical payment flow even if they cannot see the screen clearly, cannot hear an audio prompt, or cannot perform precise touch gestures. The project vision can be summarized as: one payment experience, multiple safe and accessible ways to use it.

4.2 Product Goals

- Allow payment intent to be initiated by voice.
- Support navigation by voice and keyboard.
- Read or visibly present receiver, amount, status and warnings clearly.
- Support Tamil, English and Tanglish input patterns.
- Provide different feedback modes: visual, spoken and haptic.
- Support QR scanning with voice guidance and suspicious-payload warning.
- Demonstrate secure confirmation and authorization without storing real financial secrets.
- Preserve accessibility preferences and demo data across sessions.

4.3 Non-Goals for the Current Mini-Project

- Real money transfer or production bank settlement.
- Real UPI PIN collection, storage, logging, backend transmission or read-aloud.
- Production KYC or RBI/NPCI onboarding.

- Real bank account discovery through a PSP/common library without authorized integration.
- Production fraud machine learning or risk scoring.
- Raw fingerprint or face-data collection.
- Universal phone-number-to-real-name lookup.
- Guaranteeing Gemini or network-dependent AI while fully offline.

5. Target Users and Accessibility Personas

Persona	Primary needs	Relevant capabilities
Vision / Blind	Non-visual navigation, state awareness, receiver/amount verification.	Screen reader labels, ARIA live regions, spoken feedback, voice navigation, high contrast.
Low Vision	Legible content, strong focus indicators, reduced visual ambiguity.	Large text, high contrast, clear borders, responsive layout.
Hearing / Deaf	Non-audio alternatives for every important event.	Captions, visual status, confirmation cards, vibration/flash-style feedback.
Motor / Dexterity	Reduced precision and repetition.	Large targets, dwell click, keyboard operation, fewer taps, emergency stop.
Cognitive / Elderly	Simple decisions, less clutter, predictable sequence.	Simple mode, step-by-step confirmation, plain language, reduced visual noise.
Tamil / Tanglish user	Conversational commands without formal English syntax.	Tamil/English/Tanglish parsing, local-language feedback preference.

6. Functional Scope and Core Features

ID	Feature	Requirement / behaviour	Status
F01	Voice Payment	Interpret a spoken payment request, extract recipient and amount, validate them and move to explicit confirmation.	Current / integrated
F02	Voice Navigation	Support intents such as Home, History, Back, Balance and QR help.	PRD requirement; natural assistant support integrated
F03	Audio Feedback	Read important states aloud using browser TTS/speechSynthesis in the current build.	Current / integrated
F04	Voice-Guided QR Scanner	Open camera, guide positioning, decode QR, parse payment fields and warn on suspicious payloads.	Current QR scanner integrated; guidance is part of accessibility flow

F05	Strict Confirmation Gate	Read back receiver and amount; require explicit Yes/No before simulated execution.	Core safety requirement
F06	Haptic Feedback	Use device/browser vibration patterns for confirmation, success and warning when supported.	Current accessibility capability
F07	Screen Reader Support	Use semantic HTML, labels, focus management and ARIA live announcements.	Current / integrated
F08	Balance and History	Show/read mock balance, append simulated transactions and support transaction details/search.	Current / integrated
F09	Bank Linking Demo	Guided 6-step simulated bank linking, primary bank choice and masked account data.	Current / integrated simulation
F10	Secure Authorization	Use WebAuthn/platform authenticator where supported; use clearly labelled demo fallback otherwise.	Current / integrated
F11	Beneficiary Management	Add, edit, delete, favorite, quick pay and voice search contacts.	Current / integrated
F12	Bills and Recharge	Provide mock bill/recharge interaction from quick actions / voice where implemented.	MVP feature / mock flow
F13	Accessible Settings	Theme, language, accessibility modes, voice/speech, bank and security settings.	Current / integrated
F14	Receipts	Display transaction detail, read receipt and share receipt text.	Current / integrated
F15	PWA / Offline Shell	Cache app shell and preferences; network-dependent APIs fall back when unavailable.	Current / integrated
F16	Fraud and Safety Guards	Warn for wrong receiver, suspicious QR, invalid amount, duplicate attempts and high amount scenarios.	Current / integrated simulated guards

7. User Stories and Primary Use Cases

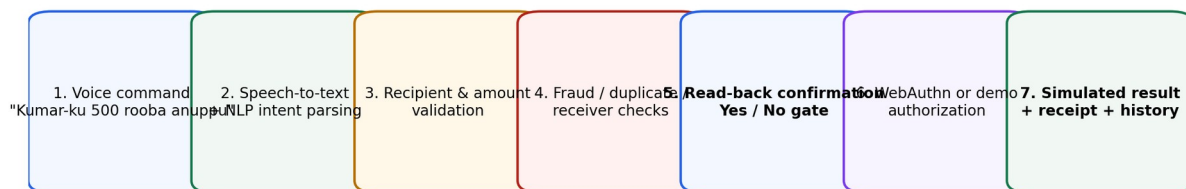
- As a blind user, I want the receiver and amount read aloud before confirmation so that I can verify the transaction without relying on the screen.

- As a hard-of-hearing user, I want every spoken message to have a visible equivalent so that audio is not required.
- As a user with limited dexterity, I want large controls and keyboard/dwell alternatives so that I do not need precise taps.
- As a Tamil or Tanglish speaker, I want to say commands naturally, for example "Kumar-ku 500 rooba anuppu," instead of learning rigid English commands.
- As a safety-conscious user, I want unknown receivers and suspicious QR codes clearly flagged before confirmation.
- As a user, I want to cancel at the confirmation stage without any debit or partial state change.
- As a user, I want successful simulated transactions to appear in history and update the mock balance consistently.

8. Main User Flows

8.1 Voice Payment Flow

Voice Payment - End-to-End Flow



Safety principle: AI may interpret intent, but deterministic rules and explicit user confirmation control the payment simulation.

Figure 1. Deterministic safety gates remain separate from AI intent understanding.

8.2 QR Scan & Pay Flow

QR Scan & Pay Flow



Known payees can follow a verified-demo path. Unknown or suspicious payloads remain explicitly marked and require extra caution.

Figure 2. QR data is decoded and verified before the user reaches confirmation.

8.3 Balance Flow

User asks for balance -> application identifies the selected mock bank account -> optional secure confirmation if required -> mock balance is displayed visually and read aloud -> no financial credential is exposed.

8.4 Transaction Search Flow

User requests a recent transaction or a beneficiary name -> transaction history is filtered -> results appear in an accessible list -> optional voice readout is provided -> transaction details can be opened by click or keyboard.

9. Accessibility Requirements

ID	Requirement	Specification
A11Y-01	Semantic structure	Critical controls must have programmatic names and roles; use semantic HTML before custom ARIA.
A11Y-02	Screen reader feedback	Dynamic confirmations, warnings and payment results should use appropriate live announcements.
A11Y-03	Keyboard-only operation	Core flows must be operable without a pointer device.
A11Y-04	Large targets	Critical buttons should be large enough

		for motor-access needs and touch use.
A11Y-05	High contrast	Provide strong foreground/background separation, borders and focus indicators.
A11Y-06	Large text	Allow persistent text-size preference without clipping or loss of information.
A11Y-07	Captions	Spoken prompts should have synchronized or equivalent visible text.
A11Y-08	Haptic alternatives	Use vibration where available, but never make vibration the only method of communicating a critical state.
A11Y-09	Multimodal redundancy	Critical actions must not depend on one sense only.
A11Y-10	Cognitive simplicity	Use direct wording, predictable sequencing and reduced visual noise.
A11Y-11	Language	Support English and Tamil output preferences and Tanglish input understanding.
A11Y-12	Responsive use	Support mobile widths around 320-430 px and larger tablet/desktop layouts without loss of functionality.

WCAG wording note

The project materials reference WCAG-friendly design and, in the user description, "WCAG 2.1 AAA" for the vision mode. A formal AAA compliance claim should only be made after a complete conformance audit across all applicable success criteria; this report therefore describes the implemented accessibility techniques rather than certifying compliance.

10. AI, Voice and NLP Requirements

Dex is the conversational assistant. Gemini is used as an intent-understanding layer rather than as the payment executor. The current feature summary reports Gemini-backed natural communication through the backend and gives examples such as `balance_check`, `payment_request`, `transaction_history` and `scan_qr_help` intents.

10.1 AI Responsibilities

- Understand natural Tamil, English and Tanglish user input.
- Identify high-level intent such as payment, balance, history or QR help.
- Extract safe fields such as recipient and amount when present.
- Generate conversational assistance and guidance.
- Use the active profile name naturally in responses.

10.2 AI Non-Responsibilities / Guardrails

- Do not execute a payment.
- Do not bypass receiver verification, fraud checks, confirmation or authorization.
- Do not receive or store a real UPI PIN, OTP or raw biometric material.
- Do not expose the Gemini API key in frontend code.
- Do not treat model output as trusted transaction state without deterministic validation.

10.3 Current Speech Output

The current integrated-features document states that spoken output uses browser TTS / speechSynthesis fallback. Gemini Tamil TTS is listed as not yet confirmed. The broader project description mentions gTTS; if gTTS is not present in the current codebase, it should be documented as a proposed alternative rather than an implemented dependency.

11. Security, Privacy and Fraud-Safety Requirements

ID	Control	Requirement
SEC-01	UPI PIN prohibition	Never collect, store, log, send to backend, send to Gemini, or read aloud a real UPI PIN.
SEC-02	OTP prohibition	Real OTP is not part of the current bank-linking simulation; do not persist OTP secrets.
SEC-03	Mock execution	Current financial execution must remain simulated.
SEC-04	Explicit confirmation	Voice intent is not authorization. Receiver and amount must be presented before a Yes/No gate.
SEC-05	Authorization	Use WebAuthn/platform authenticator where supported; label demo fallback clearly.
SEC-06	Duplicate protection	Prevent double processing / double-debit simulation through a processing lock and duplicate detection.
SEC-07	Receiver warning	Flag ambiguous, wrong, unknown or unverified beneficiaries.
SEC-08	QR safety	Block or warn on suspicious URL/QR payloads before payment review.
SEC-09	Amount validation	Reject zero/negative amounts and support high-amount / limit warnings.
SEC-10	Secrets	Store Gemini API key on the backend in environment configuration, not in the

		browser.
SEC-11	Audit hygiene	If events are logged, mask sensitive data and record safe events only.
SEC-12	Emergency stop	Provide a quick cancel/help control that returns to a safe state.

Security Design Principle

AI interprets intent; deterministic application logic controls transaction state. This separation is central to preventing a generative response from becoming a payment authorization.

12. System Architecture

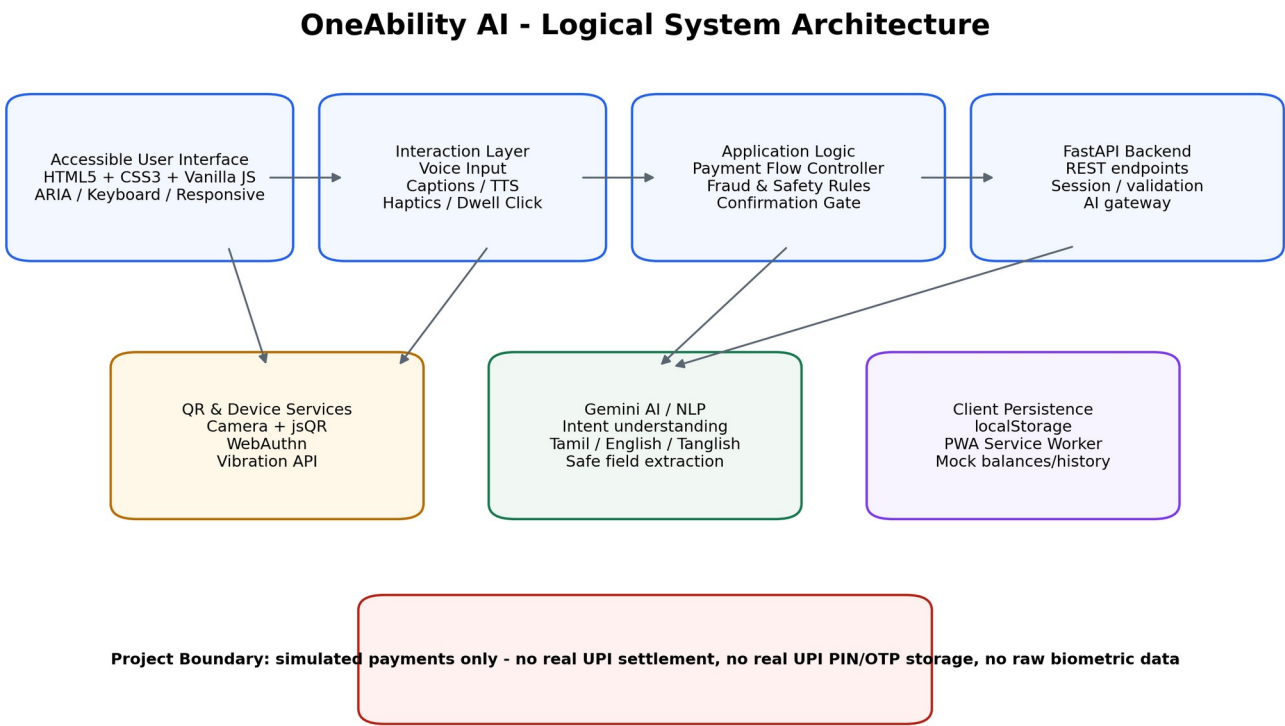


Figure 3. Logical architecture for the current simulated OneAbility AI system.

12.1 Component Responsibilities

Layer / Component	Responsibilities
Accessible Presentation Layer	Responsive dashboard, semantic controls, accessible settings, transaction cards, captions, focus states and confirmation screens.
Interaction Layer	Voice capture, browser speech output, haptics, keyboard/dwell alternatives and multimodal feedback orchestration.

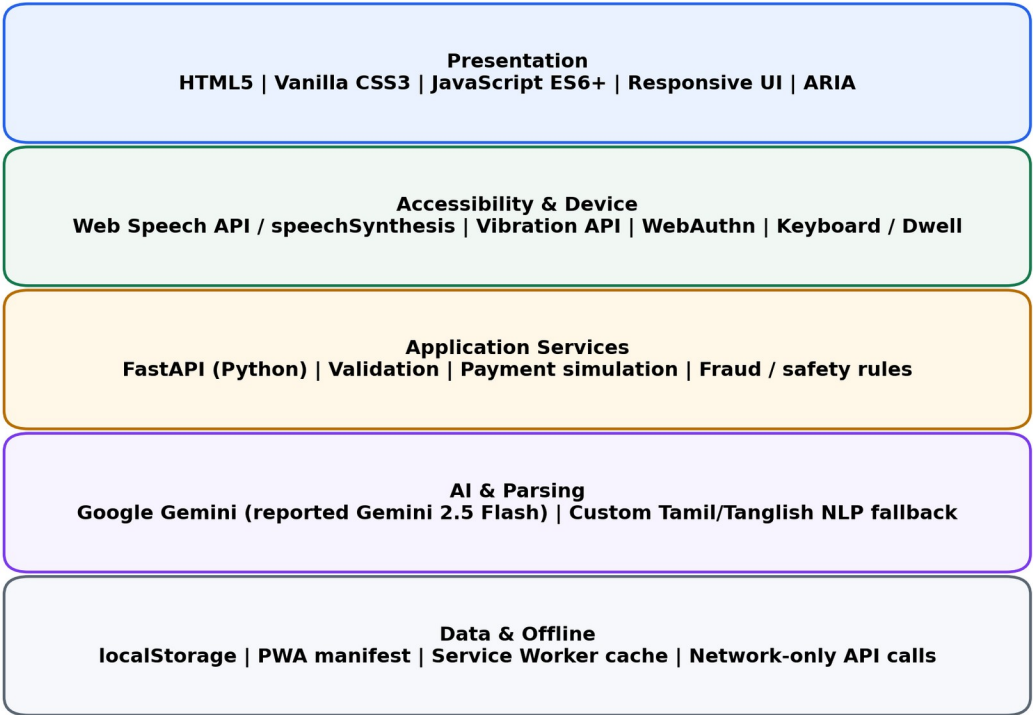
Application Logic	Payment state machine, explicit confirmation, validation, mock balance/history updates, duplicate lock and emergency cancel.
QR / Device Services	Camera access, jsQR decoding, WebAuthn secure credential prompt, vibration capability.
FastAPI Backend	REST endpoints for AI chat, QR verification, voice/payment support routes as currently implemented, plus health/session validation.
Gemini / NLP	Natural-language understanding and conversational support; no direct control of financial state.
Client Persistence	localStorage for profile, mock banks, balance, beneficiaries, settings and transaction history; service worker for app-shell cache.

12.2 Trust Boundaries

- Browser boundary: UI, camera, WebAuthn and localStorage run on the client device.
- Backend boundary: Gemini credentials and backend validation remain server-side.
- AI boundary: model output is untrusted input to deterministic application logic.
- Financial boundary: no real bank settlement or real financial secrets enter the mini-project flow.
- Offline boundary: static shell and local state can remain available; AI/API-dependent features require network or deterministic fallback.

13. Technology Stack

Technology Stack - Current Implementation View



Runtime note: a Node.js server.js may serve/orchestrate the frontend locally, while application APIs are provided by FastAPI.

Figure 4. Current implementation stack based on the supplied project documents.

Area	Technology	Purpose	Status
Frontend	HTML5, Vanilla CSS3, JavaScript ES6+	Core web UI, responsive dashboard, semantic accessibility and interaction logic.	Confirmed
Backend	Python, FastAPI, Uvicorn	REST API, AI gateway, validation, session-oriented service logic.	Confirmed
AI / NLP	Google Gemini API; reported Gemini 2.5 Flash; custom/local parsing fallback	Natural intent understanding in Tamil, English and Tanglish; safe field extraction.	Confirmed / reported
Speech Input / Output	Web Speech API; browser speechSynthesis/TTS	Voice input and current spoken feedback.	Confirmed current output
QR	Camera API + jsQR	Live QR capture and decode, followed by payload parsing/verification.	Confirmed
Biometric-style authorization	WebAuthn / platform authenticator	Secure credential flow without access to raw fingerprint/face	Confirmed

		data.	
Haptics	Navigator Vibration API	Optional vibration cues on supported browser/device.	Confirmed capability
Persistence	browser localStorage	Profile, settings, mock bank/balance, beneficiaries and transaction history.	Confirmed
Offline	PWA manifest + Service Worker	App-shell cache and offline opening after initial caching.	Confirmed
Runtime / local hosting	Node.js server.js (user description)	May serve/orchestrate frontend locally; FastAPI remains the application API backend.	Description-level; verify exact role in code

14. Data and Persistence Design

The current system does not use a centralized cloud database. Demo data is primarily kept in browser localStorage. This makes the current design appropriate for a local academic demonstration, but not for multi-device financial production use.

Logical Entity	Representative fields	Notes
UserProfile	displayName, optional nickname, language, accessibility preferences	Local identity/personalization only; not production KYC.
LinkedBankDemo	bankName, maskedAccount, primaryFlag, mockBalance	Simulated bank account state.
Beneficiary	name, UPI-like identifier, verifiedDemoFlag, favoriteFlag	Predefined or user-added demo contacts; custom entries may default to unverified.
Transaction	mockTransactionId, recipient, amount, dateTime, status, bank, method	Simulated payment history and receipt data.
AppSettings	theme, language, contrast, largeText, captions, dwell, simpleMode	Persistent accessibility and appearance configuration.
Session / ProcessingState	current flow state, confirmation state, processing lock	Prevents accidental duplicate processing and supports safe cancellation.

14.1 Data Handling Rules

- Do not store real UPI PIN, OTP or raw biometric data.
- Mask account identifiers shown in the UI.
- Treat localStorage as demo persistence, not secure financial storage.
- Keep transaction IDs explicitly labelled as mock/demo values.
- If a future cloud database is added, introduce authentication, authorization, encryption, retention rules and auditable data ownership.

15. API and Integration Design

The current integrated-features report identifies the following backend capabilities. Exact route names should be verified against the final codebase before submission because the source document notes that some earlier voice routes may have changed during refactoring.

Endpoint	Purpose	Documentation status
POST /api/assistant/chat	Gemini-backed conversational understanding and response.	Reported current
POST /api/qr/verify	QR verification flow and payee/payload validation.	Reported current
POST /api/voice/parse-payment	Earlier voice payment parsing route.	Reported earlier; verify final code
POST /api/voice/confirm-payment	Earlier voice confirmation backend route.	Reported earlier; verify final code
GET /health	Backend health check used during local testing.	Reported earlier

15.1 Example Safe Payment Payload

Illustrative only

A safe simulated payment request can contain recipient identifier, amount, selected demo bank and a client-generated flow/session ID. It should never contain a real UPI PIN, OTP or raw biometric data.

16. PWA and Offline Behaviour

The PWA manifest and service worker provide an offline application shell after caching. The static UI can open offline and local preferences remain available. API requests are network-dependent, so Gemini and backend-required functions cannot be assumed to work offline.

Scenario	Expected behaviour
App previously cached, network unavailable	Open static application shell; load local preferences/history where available.
Gemini unavailable	Use deterministic local parser fallback where supported; do not fabricate network results.
QR scan while offline	Camera/jsQR decoding may still work locally; any backend verification that requires network should be clearly unavailable or deferred.
Payment simulation offline	Only perform actions that are fully local and clearly labelled as simulation; never imply real settlement.

Network restored	Resume API-dependent assistant features; current documents do not describe a cloud sync queue.
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17. Non-Functional Requirements

ID	Quality attribute	Requirement
NFR-01	Accessibility	Core payment flows must be operable through more than one modality and should remain keyboard/screen-reader usable.
NFR-02	Responsiveness	Support mobile, tablet and desktop layouts; current report cites 320-430 px mobile widths and 768 px+ two-column layout.
NFR-03	Safety	A failed validation or cancellation must return to a safe state with no simulated debit.
NFR-04	Performance	Voice/QR interactions should provide immediate state feedback and avoid blocking the UI during network calls.
NFR-05	Resilience	If AI is unavailable, fall back to deterministic parsing or a clear error rather than silently continuing.
NFR-06	Privacy	No real financial secrets or raw biometric data may be retained by the application.
NFR-07	Maintainability	Keep AI understanding, deterministic payment logic, accessibility services and storage responsibilities separated.
NFR-08	Auditability	Warnings, confirmations and mock transaction outcomes should be reproducible and explainable during faculty demonstration.

18. Error and Recovery Flows

Failure / exception	Expected recovery
Speech not understood	Ask the user to repeat or switch to a manual/visual input path; do not guess recipient or amount.
Recipient ambiguous	Show/read candidate or warn that the receiver is not verified;

	require correction before confirmation.
Invalid amount	Reject zero/negative/non-numeric values and request a valid amount.
Suspicious QR	Stop progression, announce warning and allow cancel/rescan.
Camera permission denied	Provide an accessible explanation and demo/manual fallback if appropriate.
WebAuthn unsupported	Use clearly labelled demo authorization fallback; do not claim biometric verification occurred.
Network/Gemini failure	Use deterministic fallback where available; otherwise show a clear retry/offline message.
Duplicate submission	Processing lock should ignore/reject the second attempt and preserve a single transaction result.
User says Cancel / No	Terminate the payment flow and return to a safe screen without balance change.

19. Acceptance Criteria and Success Metrics

- Recipient and amount are correctly extracted for predefined voice test cases.
- Confirmation clearly presents or reads receiver, identifier and amount before execution.
- Yes/No confirmation is handled correctly and cancellation leaves no simulated debit.
- Main payment flow can be navigated with keyboard and screen reader.
- Deaf/hard-of-hearing users can understand critical states without audio.
- High contrast and large-text settings persist and do not break layout.
- Suspicious QR, wrong receiver, duplicate attempt and high-amount test cases produce warnings.
- Successful mock payment updates history and mock balance exactly once.
- No real UPI PIN is present in app storage, payloads, logs or AI requests.
- Faculty demo can complete the intended end-to-end scenario without crash.

19.1 Success Metrics

- Core demo flows complete without crash.
- Voice payment demonstrations work across Tamil, English and Tanglish test styles.
- Critical screens remain keyboard- and screen-reader-accessible.
- User cancellation or errors always return to a safe state.
- The project scope can be explained accurately without implying real UPI settlement.

20. Testing and Quality Status

The supplied integrated-features report records multiple phase-specific test suites. These numbers should not be combined into a single "total tests" figure because they belong to different phases and regression suites.

Reported test phase	Result
Earlier full-project audit	285 / 285 checks passed
Responsive/settings phase	94 validations passed
Gemini backend integration	13 / 13 tests passed
Personalization backend suite	18 / 18 passed
Responsive regression	25 / 25 passed

Recommended report wording

Multiple regression suites were executed across development phases, and the reported suites passed at each phase. Final submission should rerun the current codebase and record the latest results.

21. Current Scope vs Planned Roadmap

Capability	Status	Roadmap note
Gemini Tamil TTS	Not yet confirmed	Current spoken output uses browser TTS/speechSynthesis.
Monthly expense insights	Not yet confirmed	Planned summary/category/previous-month comparison.
QR review + demo PIN screen	Not yet confirmed	Planned scan -> payee/account -> amount -> bank -> review -> safe demo authorization flow.
Real OTP API	Not integrated	Bank-linking SMS/OTP remains simulated.
Cloud database	Not integrated	Current persistence is mainly localStorage.
Real UPI settlement	Intentionally not integrated	Would require authorized PSP/bank/NPCI path and regulatory compliance.
Universal phone-number real-name lookup	Not integrated	Privacy-safe alternative is app beneficiary data or user-approved device contacts.

21.1 Production Evolution Path

- Replace local demo identity with production-grade authentication and verified user/account onboarding.
- Replace mock bank linking with an authorized payment-service-provider integration.
- Move sensitive business data from localStorage to a secured backend data layer with explicit access control.

- Use provider-compliant payment authorization without exposing financial secrets to the application or AI model.
- Perform formal accessibility conformance testing on target browsers/devices and with assistive technologies.
- Add observability, fraud monitoring, privacy controls, incident response and regulatory compliance before any real-money use.

22. Risks and Mitigations

Risk	Severity	Mitigation
AI misinterprets recipient or amount	High	Never execute from AI output directly; validate fields and force explicit receiver/amount confirmation.
User assumes demo is real payment	High	Use persistent demo/simulation labels and mock transaction identifiers.
Accessibility feature unsupported in browser	Medium	Feature-detect APIs and provide equivalent alternatives; never depend on vibration or speech alone.
LocalStorage data loss	Medium	Treat data as demo-only; provide reset/export if needed; future production requires server persistence.
QR payload is malicious	High	Parse allowlisted payment fields, block suspicious URLs, warn unknown payees and require review.
Speech recognition error in noisy environment	Medium	Offer manual entry, repeat-back and correction before confirmation.
Overclaiming WCAG AAA / biometric / UPI compliance	High	Use precise wording: implemented techniques, WebAuthn authorization, simulated payment; claim compliance only after formal verification.
Network dependency for Gemini	Medium	Use deterministic fallback or clear unavailable state; do not block safe manual payment demo flows unnecessarily.

23. Recommended Demonstration Scenario

1. Open OneAbility AI and show the selected accessibility mode and language settings.
2. Say: "Kumar-ku 500 rooba anuppu."
3. Explain that speech is converted to text and the NLP layer extracts recipient and amount.
4. Show the deterministic safety checks for receiver, amount and duplicates.
5. Let Dex/app read back: receiver + amount, while the same confirmation is visible on screen.
6. Confirm with "Aama" / Yes.
7. Trigger WebAuthn if supported, or explicitly use the labelled demo authorization fallback.

8. Complete the simulated payment and demonstrate visual, audio and haptic success feedback.
9. Open the receipt and transaction history; show mock balance update.
10. Finish by explaining that no real UPI PIN or real money settlement occurred.

24. Glossary and Viva Notes

Term	Meaning in OneAbility AI
Accessibility	Designing the application so people with different sensory, motor or cognitive needs can operate it effectively.
Simulated payment	A demo transaction that updates mock data but does not transfer real money.
NLP	Natural Language Processing - understanding user language and extracting intent/fields.
NER	Named Entity Recognition - extracting items such as recipient name and amount from text.
TTS	Text-to-Speech - converting on-screen/system text into spoken audio.
WebAuthn	A web standard for secure authentication using platform authenticators or security keys without exposing raw biometric data.
PWA	Progressive Web App - a web application with installable/offline-capable features such as a manifest and service worker.
Haptic feedback	Vibration-based feedback used as an additional interaction cue.
ARIA	Accessible Rich Internet Applications attributes used to communicate semantics and dynamic state to assistive technologies.
Deterministic safety logic	Explicit program rules whose result is predictable and testable, unlike generative AI output.
PSP	Payment Service Provider - an authorized payment integration partner required for real payment processing.
UPI	Unified Payments Interface. Real UPI settlement is intentionally outside the current mini-project scope.

24.1 Viva-Ready Answers

Is this real UPI?

No. It is a secure college simulation. Real UPI PIN and real money settlement are intentionally not

handled. The project demonstrates accessible payment UX, QR parsing, fraud checks, authorization flow and AI assistance.

What is the main innovation?

The main highlight is not only voice payment. OneAbility AI adapts the payment experience for vision, hearing, motor and cognitive accessibility needs while keeping confirmation and safety logic separate from the AI model.

Why use Gemini?

Gemini improves natural-language understanding for Tamil, English and Tanglish. It interprets user intent and safe fields, while deterministic application logic controls validation, confirmation and simulated transaction state.

Appendix A - Requirements Traceability Summary

Goal	Requirement mapping	Evidence / design mechanism
Independent accessible payment	F01-F16, A11Y-01 to A11Y-12	Voice/QR/contacts/history + multimodal feedback
No unconfirmed payment	F05, SEC-04	Explicit read-back + Yes/No confirmation gate
Multilingual interaction	F01/F02, A11Y-11	Tamil, English, Tanglish intent understanding
Safe QR interaction	F04, SEC-08	Decode, parse, verify/warn before confirmation
Secure authorization	F10, SEC-05	WebAuthn or labelled demo fallback
No sensitive credential storage	SEC-01/02/10	No PIN/OTP/raw biometric; backend-held API key
Offline accessibility shell	F15, NFR-05	PWA cache + deterministic fallback where feasible
Demo auditability	NFR-08, Section 20	Explicit mock identifiers and phase-specific testing records

Appendix B - Final Implementation Verification Checklist

- Verify the exact FastAPI route list against the final source code before submission.
- Confirm whether Node.js server.js only serves the frontend or also performs orchestration; describe its role exactly.

- Confirm the exact Gemini model name configured in the final backend environment.
- Confirm whether gTTS exists in the codebase; if not, document browser speechSynthesis as the current TTS method.
- Confirm which fraud checks are active in the current build and which are demonstration hooks.
- Rerun the latest regression tests and record new results rather than relying only on historical phase reports.
- Run a final accessibility review with keyboard, screen reader, high contrast, large text and captions.
- Ensure every screen that can resemble a real payment clearly communicates the simulation boundary.